The six phases of the SDLC and the work that would happen in RP-Cloud Labs are:

No.	SDLC Phase	Work in RP-Cloud Labs
1	Planning	Identify the project goals, budget, timeline, team members, and resources needed to develop the Tool-Session Service.
2	Requirements Analysis	Gather requirements from students, lecturers, college administrators, and RP IT staff. Determine how students request tool sessions and how the system checks account validity and license availability.
3	System Design	Design the system architecture, database, APIs, and user interface. Plan how Docker, Kubernetes, Redis, and Kafka will support the service.
4	Implementation (Coding)	Develop the Tool-Session Service, including session requests, account validation, and license-capacity checking. Store the code in GitHub and use feature branches and pull requests.
5	Testing	Test the service to verify that session requests work correctly, license limits are respected, and errors are handled. Use automated tests and GitHub Actions.
6	Deployment and Maintenance	Package the service using Docker, deploy it to Kubernetes, monitor its performance, fix bugs, and release updates when requirements change.

Conclusion: These six phases help the RP-Cloud Labs team plan, develop, test, deploy, and maintain a reliable Tool-Session Service for Rwanda Polytechnic.